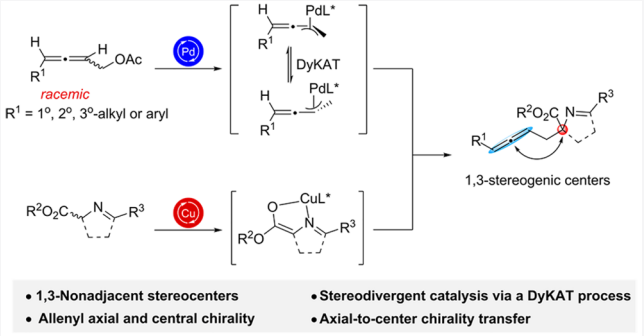
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Enantio- and Diastereodivergent Construction of 1,3-Nonadjacent Stereocenters Bearing Axial and Central Chirality through Synergistic Pd/Cu Catalysis

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ABSTRACT: In contrast to the widely explored methods for the asymmetric synthesis of molecules bearing a single stereocenter or adjacent stereocenters, the concurrent construction of 1,3-stereogenic centers in an enantio- and diastereoselective manner remains a challenge, especially in acyclic systems. Herein, we report an enantio- and diastereodivergent construction of 1,3-nonadjacent stereocenters bearing allenyl axial and central chirality through synergistic Pd/Cu-catalyzed dynamic kinetic asymmetric allenylation with racemic allenylic esters. The protocol is suitable for a wide range of substrates including the challenging allenylic esters with less sterically bulky substituents and provided chiral allenylic products bearing 1,3-nonadjacent stereocenters with high levels of enantio- and diastereoselectivities (up to >20:1 dr and >99% ee). Furthermore, several representative transformations involving axial-to-central chirality transfer were conducted, affording useful structural motifs containing nonadjacent stereocenters in a diastereodivergent manner.



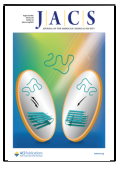
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INTRODUCTION

The synthesis of chiral molecules in enantiomerically pure form is a prominent theme in organic synthesis. A large number of methods have been developed for the preparation of chiral molecules containing a single stereocenter.[1] In addition, the asymmetric synthesis of molecules bearing adjacent stereocenters, including the challenging unfavored diastereomers, has also been realized with efficient enantio- and diastereocontrol by adjusting the structural properties of chiral catalysts.[2] However, the concurrent construction of 1,3-stereogenic centers in an enantio- and diastereoselective manner is still challenging, especially for the acyclic systems and cyclic systems in which two 1,3-stereogenic centers are not in the same ring (Scheme 1A). In classic single catalyst modes, the chiral catalyst is responsible for the co-constructions of the two nonadjacent stereogenic centers, thus is of limitation for the divergent syntheses of all the four stereoisomers. In principle, much more complex chiral catalysts possessing elaborate chiral skeletons, electronic, steric, or directing effects are required in order to achieve satisfactory stereocontrol of nonadjacent stereocenters. Thus, only a limited number of catalytic systems for the asymmetric construction of 1,3-nonadjacent stereocenters have been disclosed.[3] In view of the structural motifs containing nonadjacent stereocenters widely present in numerous natural products and bioactive compounds, the development of a simple catalytic system or strategy for the enantio- and diastereoselective construction of molecules containing nonadjacent stereocenters is highly desired, which might change the manner in which scientists think about designing and building molecules.[4]

Allene chemistry has become an important branch in organic chemistry due to the unique structural feature of the allene moiety, that is, two cumulated double bonds allowing for axial chirality.[5] Consequently, synthetic organic chemists have long pursued general and efficient methods for the catalytic asymmetric synthesis of chiral allenes.[6−9] Indeed, significant progress has been achieved in the asymmetric construction of single allenyl axial chirality.[6−8] Furthermore, adjacent stereocenters bearing allenyl axial chirality and central chirality could also be efficiently constructed through metal- or organo-catalyzed approaches.[9] However, the concurrent construction of nonadjacent stereocenters in this area remains a long-standing challenge, and no stereodivergent example has been reported yet (Scheme 1B).[3g] Importantly, nonadjacent stereocenters bearing allenyl axial and central chirality are frequently embedded in diverse bioactive molecules[5] and also provide an important method for preparing two nonadjacent central

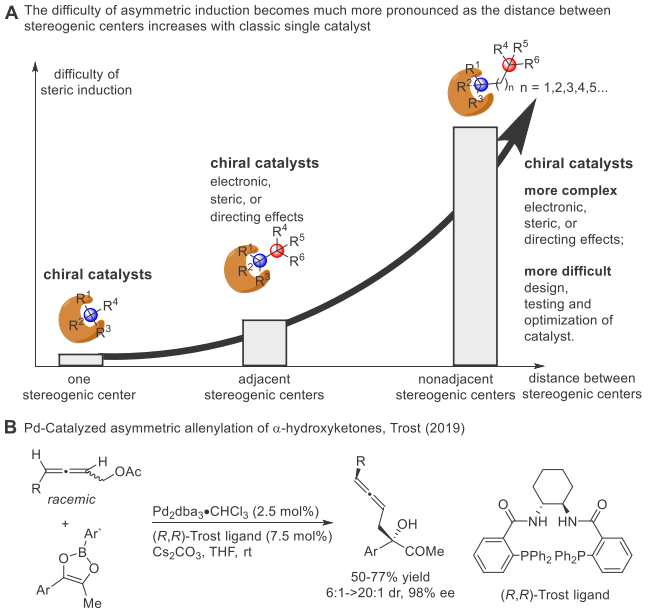
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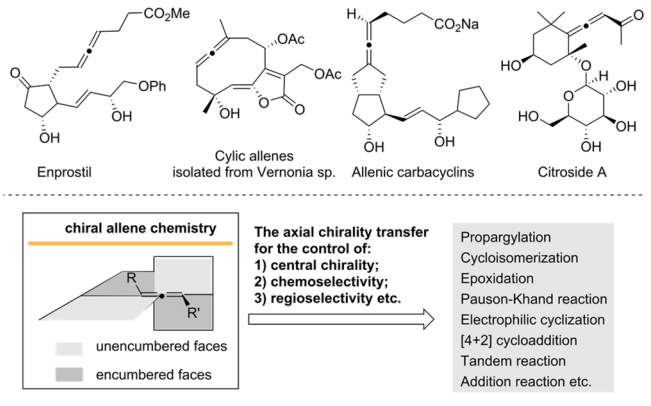
https://doi.org/10.1021/jacs.1c05087
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Scheme 1. Enantio- and Diastereodivergent Construction of 1,3-Nonadjacent Stereocenters Bearing Axial and Central Chirality



stereocenters via efficient axial-to-central chirality transfer (Scheme 2).

Scheme 2. Representative Bioactive Molecules Bearing Chiral Allene Motifs and the Chiral Allene Chemistry

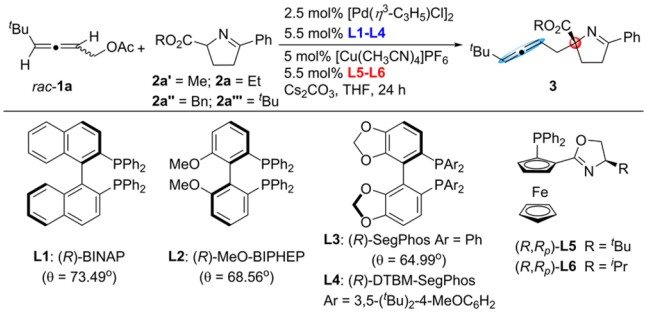


Stereodivergent syntheses are gaining increasing attention as the stereochemistry of the chiral molecules has an important influence on their bioactivity.[10−14] Dual catalysis has become one of the most powerful methods for stereodivergent synthesis, and many types of motifs bearing adjacent stereocenters have been efficiently synthesized in an enantio- and diastereodivergent manner.[12−15] To the best of our knowledge, stereodivergent dual catalysis for the construction of nonadjacent stereocenters has not been reported yet.[16] Therefore, in continuation of our interest in stereodivergent bimetallic catalysis[13a,b,e,i,l] and allene chemistry,[6d,e,g,h,8d,g−i] we envisioned bimetallic catalysis would provide a general platform for the enantio- and diastereodivergent construction of nonadjacent stereocenters (allenyl axial chirality and central chirality). It is expected that all four stereoisomers of a desired product bearing allenyl axial chirality and nonadjacent central chirality could be unprecedentedly obtained if each catalyst allows for full control over the configuration of the respective stereocenter (Scheme 1C).

RESULTS AND DISCUSSION

Initially, racemic allenylic acetate **1a** was selected as the model substrate for the dynamic kinetic asymmetric transformation (DyKAT)[17] with imino esters[18] in the presence of Cs_2CO_3 using a [Pd/**L** + Cu/**L**] catalyst combination (Table 1). At first, several classic biphosphine ligands were screened to explore the effect of the chiral ligands on the Pd catalyst in this reaction, and the product (*R*,*R*)-**3aa′** was obtained in >99% enantioselectivity with seemingly irregular yields and diastereoselectivities (entries 1−3). Careful analysis suggested that as

Table 1. Optimization of the Reaction Conditions[a]



entry	2	R	L for Pd	L for Cu	*T* (°C)	yield (%)[b]	dr[c]	ee (%)[c] (config.)
1	**2a′**	Me	**L1**	**L5**	0	88	1:1	>99 (*R*,*R*)
2	**2a′**	Me	**L2**	**L5**	0	92	1:1	>99 (*R*,*R*)
3	**2a′**	Me	**L3**	**L5**	0	39	15:1	>99 (*R*,*R*)
4	**2a′**	Me	**L4**	**L5**	0	48	1:6	>99 (*R*,*S*)
5	**2a′**	Me	**L3**	**L6**	0	55	11:1	>99 (*R*,*R*)
6	**2a′**	Me	**L3**	**L6**	−10	50	>20:1	>99 (*R*,*R*)
7	**2a**	Et	**L3**	**L6**	−10	80	>20:1	>99 (*R*,*R*)
8	**2a″**	Bn	**L3**	**L6**	−10	62	12:1	>99 (*R*,*R*)
9	**2a‴**	tBu	**L3**	**L6**	−10	86	13:1	>99 (*R*,*R*)
10[d]	**2a**	Et	**L3**	–	−10	14	1:2	>99 (*R*,*S*)
11[e]	**2a**	Et	–	**L6**	−10	nr	–	–
12[f]	**2a**	Et	**L3**	**L6**	−10	nr	–	–

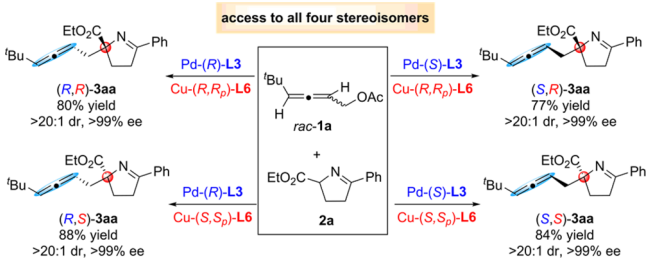
[a]Reaction conditions: **1a** (0.30 mmol, 1.5 equiv), **2** (0.20 mmol, 1.0 equiv), [Pd(η^3-allyl)Cl]$_2$ (2.5 mol %), **L1**−**L4** (5.5 mol %), [Cu(CH$_3$CN)$_4$]PF$_6$ (5.0 mol %), **L5**−**L6** (5.5 mol %), Cs_2CO_3 (0.24 mmol, 1.2 equiv), 24 h. [b]Isolated yield of all diastereoisomers; nr = no reaction. [c]Determined by HPLC analysis. [d]Without Cu catalyst. [e]Without Pd catalyst. [f]Without base Cs_2CO_3.

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the dihedral angle in the chiral backbone decreases [(*R*)-BINAP (73.49°), (*R*)-MeO-BIPHEP (68.56°), and (*R*)-SegPhos (64.99°)],[19] the diastereoselectivity increases but the yield drops by more than half. Although it is difficult to rule out the electronic effects of the ligands, the dihedral angle of the axially chiral ligands seems to play a crucial role in reactivity and diastereoselectivity. Higher diastereoselectivity was obtained with the chiral ligand possessing the smaller dihedral angle, which could exert much better asymmetric induction. Although it is difficult to understand the low reactivity of this reaction with (*R*)-SegPhos **L3**, the underlying reason might be the three-dimensional space incompatibility between the chiral Pd complex and the chiral Cu complex. Then, the ligand with higher steric hindrance (*R*)-DTBM-SegPhos **L4** was applied in this reaction. Interestingly, the complementary diastereomer (*R*,*S*)-**3aa′** was obtained in 48% yield and 1:6 dr and >99% ee (entry 4). This result suggested that the bulky steric environment of the (*R*)-DTBM-SegPhos **L4** could overwrite the stereocontrol ability of the chiral Cu catalyst and directly control both stereocenters of the allenyl esters and the nucleophiles. Subsequently, isopropyl-substituted Phosferrox (*R*,R_p)-**L6** instead of *tert*-butyl-substituted Phosferrox (*R*,R_p)-**L5** was subjected to this reaction (entry 5 vs entry 3). The yield was obviously improved with comparable enantioselectivity and diastereoselectivity. Furthermore, the diastereoselectivity increased rapidly when the reaction temperature was decreased to −10 °C (entry 6). To our delight, the imino esters substituted with Et, Bn, and ^{t}Bu are also compatible with this dual catalysis system, providing the corresponding products in high yields and with high to excellent diastereoselectivities and enantioselectivities (entries 7−9, for **2a**, 80% yield, >20:1 dr, >99% ee). Subsequently, control experiments were conducted to gain insight into the cooperative interplay of the bimetallic catalytic system (entries 10−12). The substituted product was obtained in only 14% yield in the absence of the Cu catalyst and the dr decreased rapidly from >20:1 to 1:2. No reaction occurred in the absence of the Pd catalyst. These results suggested that the combined use of two chiral metal catalysts appears to be important for the reactivity and stereochemical control of this reaction.

After establishing the feasibility of an enantio- and diastereoselective synthesis of 1,3-nonadjacent axial and central stereocenters, we then speculated if the present Pd/Cu catalytic system could furnish products with full control over the absolute and relative configuration of both axial and central chirality (Scheme 3). Indeed, by simply changing (*R*,R_p)-**L6**, the ligand for Cu, to (*S*,S_p)-**L6** while keeping the same enantiomer of (*R*)-SegPhos **L3**, (*R*,*S*)-**3aa** was formed successfully in high yield with excellent diastereoselectivity (>20:1 dr) and enantioselectivity (>99% ee). Next, the generality of enantio- and diastereodivergency was demonstrated by the synthesis of all four stereoisomers of **3aa** using all four combinations of SegPhos **L3** and Phosferrox **L6**. Gratifyingly, almost perfect stereoselectivities (>20:1 dr and >99% ee) with high reactivity were achieved using the same set of starting materials and under identical reaction conditions, which verified the efficiency of the stereodivergent bimetallic catalysis for the stereocontrol of the 1,3-nonadjacent stereocenters.

Scheme 3. Synthesis of All Four Stereoisomers of 3aa

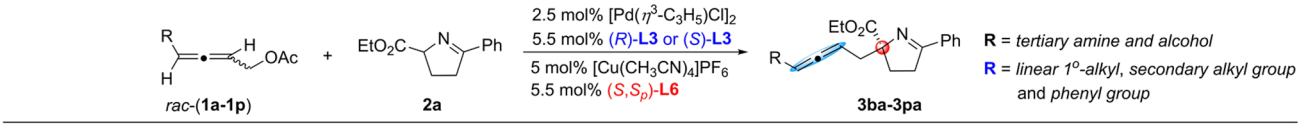


Having established the optimized reaction conditions, we explored the specific performance of our bimetallic catalytic system in the DyKAT of racemic allenylic acetates in a diastereodivergent manner (Table 2). For most of the substrates, there is no obvious difference in catalytic activity and stereoinduction of the two catalyst combinations. First, for the substrates bearing the synthetically attractive tertiary amine and alcohol groups (**1b**−**1h**), the substitution reaction proceeded smoothly to afford the desired products (**3ba**-**3ha**) with 1,3-nonadjacent axial and central chirality in high yields, and with excellent diastereoselectivities (10:1 to >20:1 dr) and enantioselectivities (96% to >99% ee). In general, sterically bulky substituent of 1,3-disubstituted allenes together with the elaborate sterically hindered ligand DTBM-SegPhos are indispensable for achieving high enantioselectivity due to the loosening nature of the allene axial chirality spreading over three carbon atoms.[6d,e,g] Accordingly, we were very curious whether our bimetallic catalytic system is suitable for substrates substituted with much less sterically bulky group. To our delight, the allenylic esters bearing a linear 1°-alkyl group (**1i**−**1n**) reacted with imine ester to afford products (**3ia**−**3na**) with high diastereoselectivities (up to 9:1 dr) and excellent enantioselectivities (up to >99% ee). Furthermore, the substrates with a secondary alkyl group (**1o**) afforded the desired products (*R*,*S*)-**3oa** and (*S*,*S*)-**3oa** with moderate diastereoselectivities (12:1 dr and 4:1 dr, respectively) and complete enantiocontrol. To achieve satisfactory diastereoselectivity of the products (*R*,*S*)-**3ia**−**3oa**, Pd/(*R*)-DTBM-SegPhos and Cu/(*S*,S_p)-**L6** were employed. However, the three-dimensional space incompatibility between the Pd/(*R*)-DTBM-SegPhos and the Cu/(*S*,S_p)-**L6** led to relatively low yields. Additionally, **1p** bearing a planar phenyl group is also compatible with this Pd/Cu dual catalytic system, giving the corresponding products (*R*,*S*)-**3pa** and (*S*,*S*)-**3pa** with satisfactory results (82% yield, >20:1 dr, >99% ee and 50% yield, 9:1 dr, >99% ee).

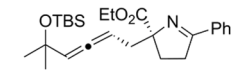
Next, the scope of the nucleophile was investigated in a diastereodivergent manner to explore the generality and limitations of our dual catalytic system. As described in Table 3, the influence of the substituents on the aromatic ring was initially investigated, and it was apparent that the position and electronics of the substituted groups exerted a limited effect, affording the desired products in high yields with excellent stereoselectivities (up to >20:1 dr and >99% ee). For example, both electron-donating and electron-withdrawing groups at the *para* position of the phenyl ring were well tolerated and delivered the corresponding products **3ab**−**3ae** in high yields, with 12:1 to >20:1 dr and >99% ee. As shown by examples **3af**−**3aj**, substituents at the *meta* and *ortho* positions were also compatible as well. Furthermore, reactions involving naphthyl-, piperonyl, and furyl-substituted substrates also proceeded well to deliver their desired products (**3ak**−**3am**) in high yields and with excellent stereoselectivities. The

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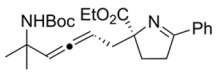
Table 2. Enantio- and Diastereodivergent Allenylation with Allenylic Acetates[a]



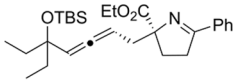
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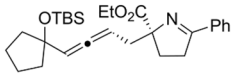
(*R*,*S*)-**3ba**
63% yield
>20:1 dr, >99% ee



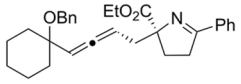
(*R*,*S*)-**3ca**
66% yield
>20:1 dr, >99% ee



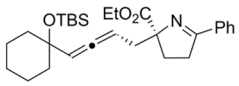
(*R*,*S*)-**3da**
73% yield
14:1 dr, >99% ee



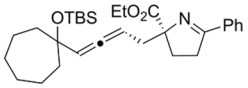
(*R*,*S*)-**3ea**
73% yield
10:1 dr, >99% ee



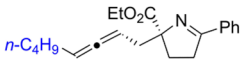
(*R*,*S*)-**3fa**
79% yield
20:1 dr, 98% ee



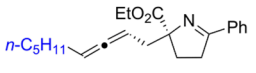
(*R*,*S*)-**3ga**
94% yield
>20:1 dr, >99% ee



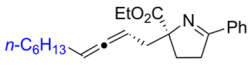
(*R*,*S*)-**3ha**
82% yield
>20:1 dr, 96% ee



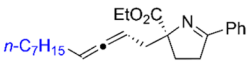
(*R*,*S*)-**3ia**[b]
48% yield
6:1 dr, >99% ee



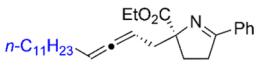
(*R*,*S*)-**3ja**[b]
52% yield
5:1 dr, 98% ee



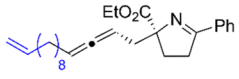
(*R*,*S*)-**3ka**[b]
62% yield
7:1 dr, >99% ee



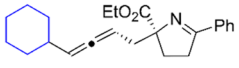
(*R*,*S*)-**3la**[b]
43% yield
9:1 dr, >99% ee



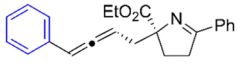
(*R*,*S*)-**3ma**[b]
83% yield
5:1 dr, >99% ee



(*R*,*S*)-**3na**[b]
81% yield
5:1 dr, >99% ee

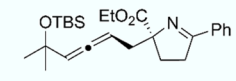


(*R*,*S*)-**3oa**[b]
51% yield
12:1 dr, >99% ee

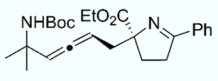


(*R*,*S*)-**3pa**
82% yield
>20:1 dr, >99% ee

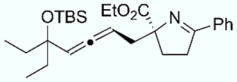
For (*S*,*S*)-**3**



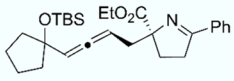
(*S*,*S*)-**3ba**
55% yield
>20:1 dr, >99% ee



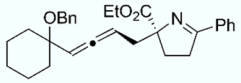
(*S*,*S*)-**3ca**
72% yield
>20:1 dr, >99% ee



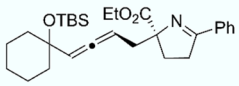
(*S*,*S*)-**3da**
80% yield
>20:1 dr, >99% ee



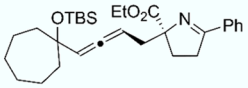
(*S*,*S*)-**3ea**
71% yield
15:1 dr, >99% ee



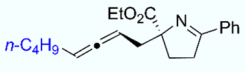
(*S*,*S*)-**3fa**
74% yield
>20:1 dr, 99% ee



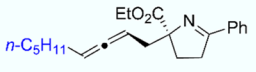
(*S*,*S*)-**3ga**
94% yield
>20:1 dr, >99% ee



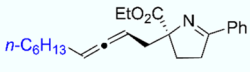
(*S*,*S*)-**3ha**
79% yield
>20:1 dr, >99% ee



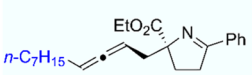
(*S*,*S*)-**3ia**
88% yield
8:1 dr, >99% ee



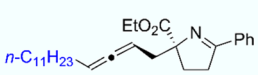
(*S*,*S*)-**3ja**
75% yield
8:1 dr, 87% ee



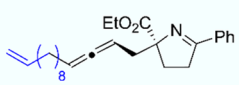
(*S*,*S*)-**3ka**
89% yield
5:1 dr, >99% ee



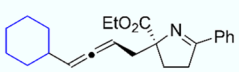
(*S*,*S*)-**3la**
77% yield
5:1 dr, >99% ee



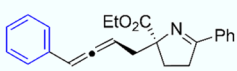
(*S*,*S*)-**3ma**
89% yield
3:1 dr, 96% ee



(*S*,*S*)-**3na**
77% yield
4:1 dr, >99% ee



(*S*,*S*)-**3oa**
87% yield
4:1 dr, 99% ee



(*S*,*S*)-**3pa**[c]
50% yield
9:1 dr, >99% ee

[a]Reaction conditions: See Table 1, entry 7. [b]Using DTBM-SegPhos instead of SegPhos. [c]Using DM-SegPhos instead of SegPhos.

absolute configurations of (*R*,*S*)-**3ak** and (*S*,*S*)-**3ak** were determined by X-ray crystallography.

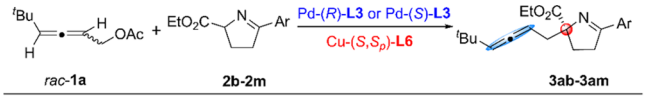
Glycine-based ketimine ester (**2n**) was also employed in our Pd/Cu bimetallic catalytic system. The desired products (*R*,*S*)-**3an** and (*S*,*S*)-**3an** were smoothly obtained in high yields and with high stereoselectivities (Table 4). Additionally, five aldimine esters (**2o**−**2s**) derived from both natural and non-natural amino acids (R = Me, Et, ^{n}Pr, allyl, and CH_2O^tBu) were utilized in our Pd/Cu bimetallic catalytic system. The allenylated amino acid products (**3ao**−**3as**) were smoothly obtained in high yields and high to excellent stereoselectivities (up to >20:1 dr and >99% ee).

Subsequently, we wondered whether these reaction conditions could be extended to the stereodivergent synthesis of all eight stereoisomers of **3qa** bearing three stereocenters (Scheme 4). Initially, when the allenylic substrate bearing a chiral secondary alcoholic group **1q** was conducted with **2a**, four of the possible stereoisomers of **3qa** were successfully obtained with this protocol by the random combinations of SegPhos **L3** and Phosferrox **L6**. The key to success might be due to the loosening feature of the axial allene chirality spreading over three carbon atoms, which leads to negligible asymmetric induction of the existing chiral alcohol motif on the axial and central chirality. Expectedly, the four remaining stereoisomers of **3qa** were expediently prepared using **1q′** and a similar protocol. Finally, all eight isomers of **3qa** were obtained in high yields with excellent stereocontrol (up to >20:1 dr).

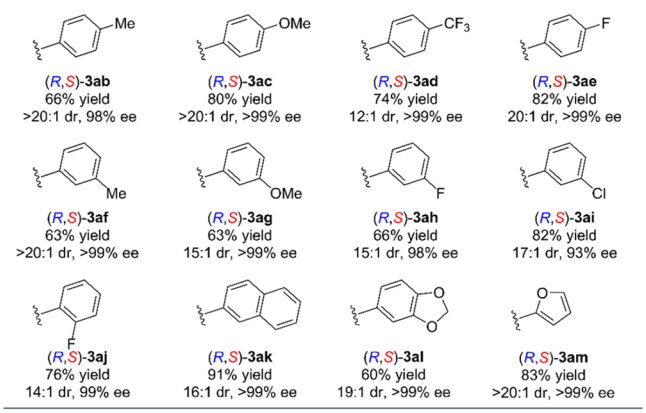
A preliminary mechanism has been proposed in Scheme 5. For the Pd catalytic cycle, the reaction is initiated with Pd^0/**L3**,

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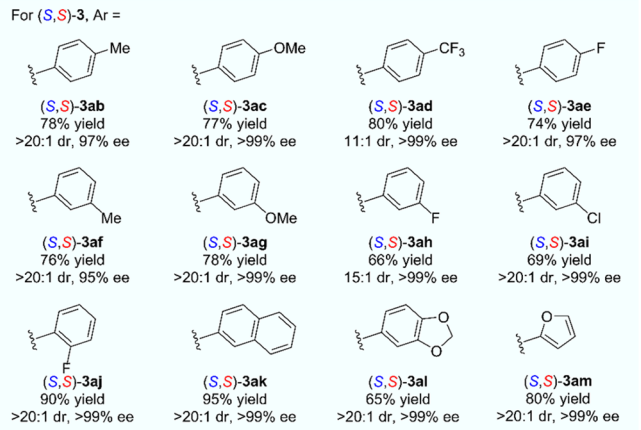
Table 3. Substrate Scope of Imino Esters[a]

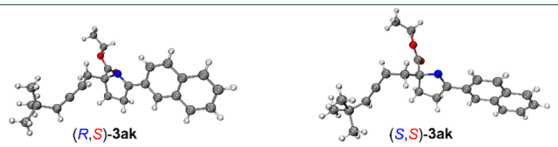


For (*R*,*S*)-**3**, Ar =



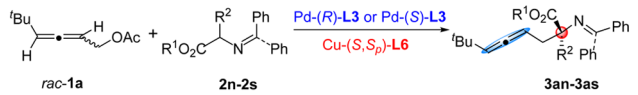
For (*S*,*S*)-**3**, Ar =

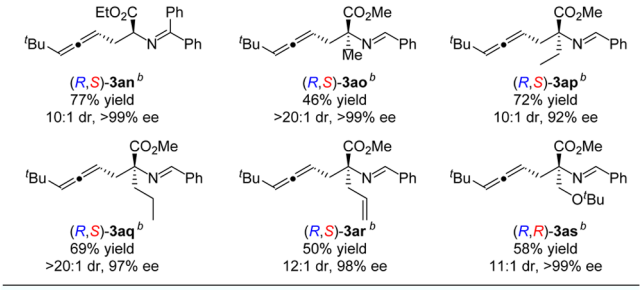


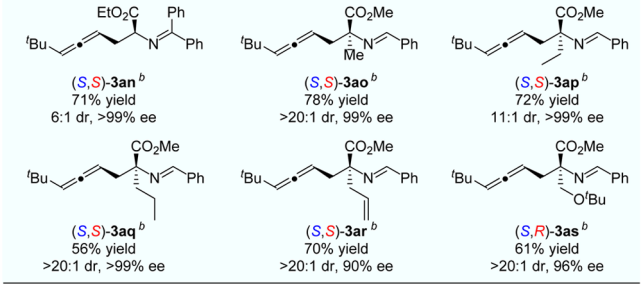


[a]Reaction conditions: See Table 1, entry 7.

Table 4. Substrate Scope of Aldimine Esters[a]

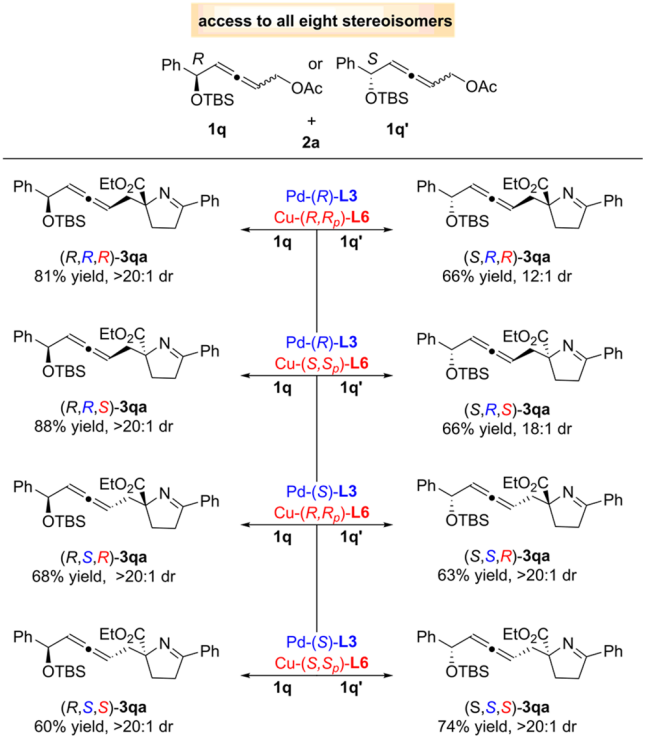






[a]Reaction conditions: See Table 1, entry 7. [b]The yield of product was determined after hydrolysis. The dr and ee values were measured after N-protecting with TsCl.

Scheme 4. Synthesis of All Eight Stereoisomers of 3qa

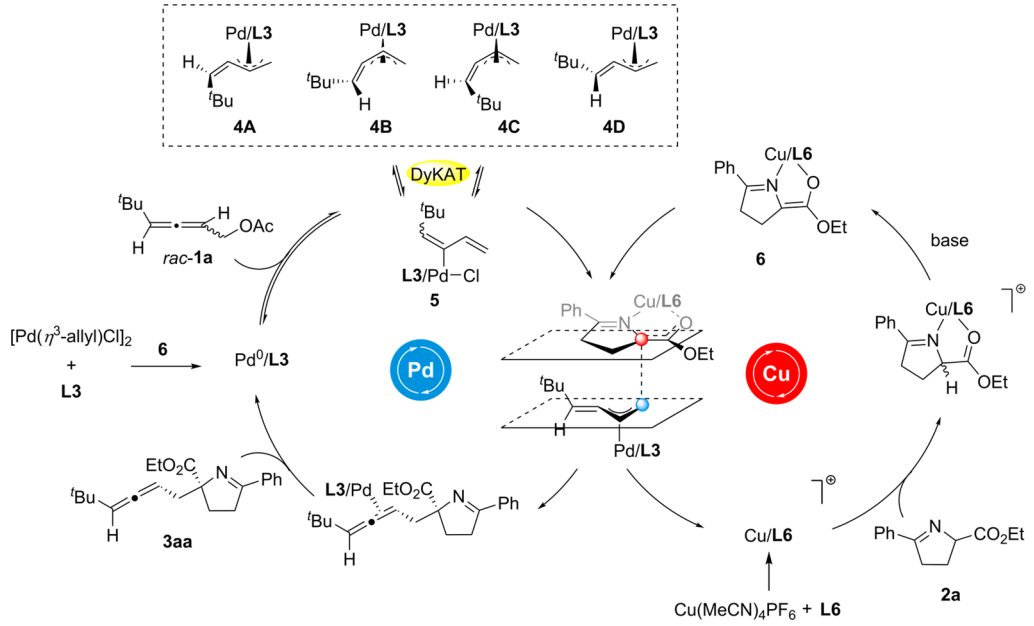


which is generated from $[Pd(allyl)Cl]_2$ via the attack of nucleophile. Subsequently the chiral Pd^0/**L3** catalyst would undergo oxidative addition reaction with racemic allenyl substrate **1a**, producing the four possible α-alkynylidene π-allylpalladium species **4A**, **4B**, **4C**, and **4D**. These four diastereomeric intermediates could interconvert via σ-(1,3-dien-2-yl)palladium **5**.[20] For the Cu catalytic cycle, the imino ester **2a** can be easily transformed into the nucleophilic N-metalated azomethine ylide **6** in the presence of a chiral Cu/**L6** catalyst and Cs_2CO_3. The nucleophile **6** then attacks the electrophilic α-alkynylidene π-allylpalladium intermediate **4** to give the corresponding product **3aa**.

In order to explore the specific performance of this bimetallic catalytic system in the DyKAT of racemic allenyl substrates, the reaction of *rac*-**1a** with **2a** was monitored at 0.5 h, 1.0 h, 3.0 h, 6.0 h, 8.0 h, and 10.0 h, and the recovered allenyl substrate **1a** was analyzed by HPLC with a chiral CHIRALPAK IC-3 column (Figure 1A). These results indicated that (*R*)-**1a** and (*S*)-**1a**[21] were consumed together at almost the same speed. Additionally, the conversion of **2a** was determined by GC-MS using mesitylene as the internal standard, and the ee and dr values of the crude products at different times were determined by HPLC with a chiral CHIRALPAK IC-3 column (Figure 1B). It was found that the enantioselectivities and diastereoselectivities of the corresponding products (*R*,*S*)-**3aa** were almost unchanged.

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Scheme 5. Proposed Mechanism of Stereodivergent Pd/Cu-Catalyzed Dynamic Kinetic Asymmetric Allenylation with Racemic Allenylic Esters



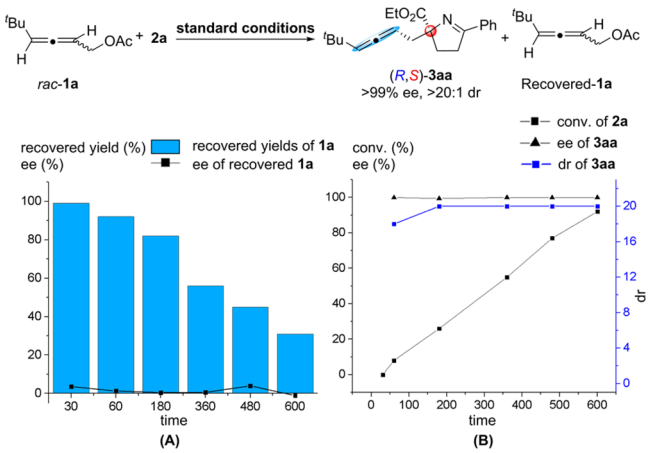
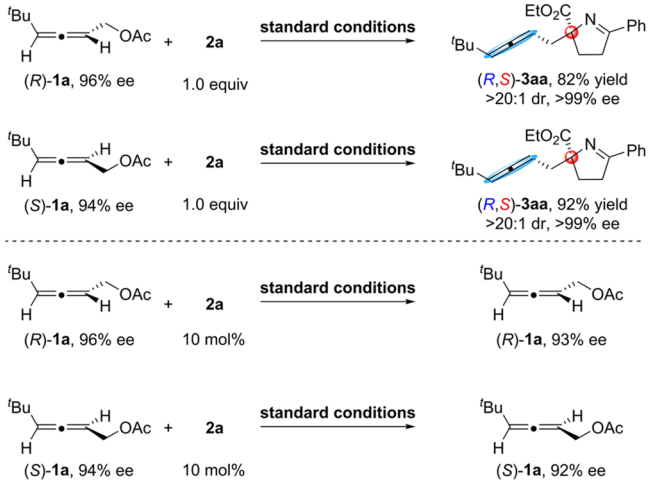


Figure 1. (A) Yields and ee values of recovered **1a** at 0.5 h, 1.0 h, 3.0 h, 6.0 h, 8.0 h, and 10.0 h. (B) The conversion of **2a** and ee and dr values of the crude products **3aa** at 0.5 h, 1.0 h, 3.0 h, 6.0 h, 8.0 h, and 10.0 h.

Scheme 6. Reactions with Enantiomerically Pure (*R*)-1a and (*S*)-1a

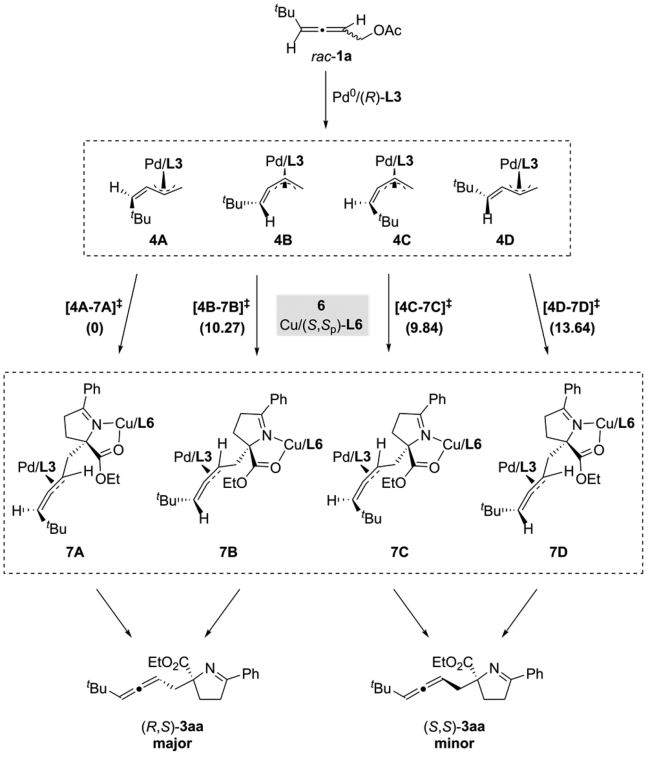


To gain an in-depth insight of this DyKAT process, reactions with enantiomerically pure (*R*)-**1a** and (*S*)-**1a** were conducted under the standard reaction conditions (Scheme 6). The same products (*R*,*S*)-**3aa** were obtained in high yields (82% vs 92% yield) and with almost the same stereoselectivities (>20:1 dr and >99% ee). Furthermore, reactions of enantiomerically pure (*R*)-**1a** and (*S*)-**1a** in the presence of 10 mol % of **2a** were also conducted under the standard reaction conditions, and no obvious racemization was found (10 mol % of **2a** is responsible for the generation of the catalytically active Pd^0 species from $[Pd(\eta^3\text{-}C_3H_5)Cl]_2$). Based on the above-mentioned results and the previous work,[6d−g] we propose that the rapid interconversion among the four α-alkynylidene π-allylpalladium intermediates (**4A**, **4B**, **4C**, and **4D**) via σ-(1,3-dien-2-yl)palladium **5** occurs before attack by the nucleophile **6** (Scheme 5).[20]

According to the Curtin−Hammett principle,[22] the ratio of products is mainly controlled by the difference ($\Delta\Delta G^{\ddagger}$) in the standard free energies of the transition states of the nucleophilic substitution step. Subsequently, theoretical calculations were conducted to investigate this process using the combinations of Pd/(*R*)-**L3** and Cu/(S,S_p)-**L6**. In the previous work, the stereocontrol of Cu/(S,S_p)-**L6** on the imino ester nucleophile has been well studied by DFT calculations.[13l,23] Thus, four possible nucleophilic substitution pathways were established (Scheme 7).[24] The calculations indicated that the relative energies of these four transition states [**4A**−**7A**]‡, [**4B**−**7B**]‡, [**4C**−**7C**]‡, and [**4D**−**7D**]‡ are 0.0, 10.27, 9.84, and 13.64 kcal, respectively. The experiments and calculations suggested that *rac*-**1a** first reacts with the Pd^0/**L3** generated in situ, affording the four α-alkynylidene π-allylpalladium intermediates **4A**, **4B**, **4C**, and **4D**, which could interconvert rapidly. Next, the nucleophile **6** attacks the electrophilic π-allylpalladium intermediates, affording the

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Scheme 7. Study of Four Reaction Pathways of *rac*-1a with 2a Using the Combinations of Pd/(*R*)-L3 and Cu/(S,S_p)-L6 by Theoretical Calculations



major product (*R*,*S*)-**3aa** from the intermediate **4A** via the transition state [**4A**−**7A**]‡ with the lowest Gibbs free energy. These results are in good agreement with the experimental observations (Figure 2).

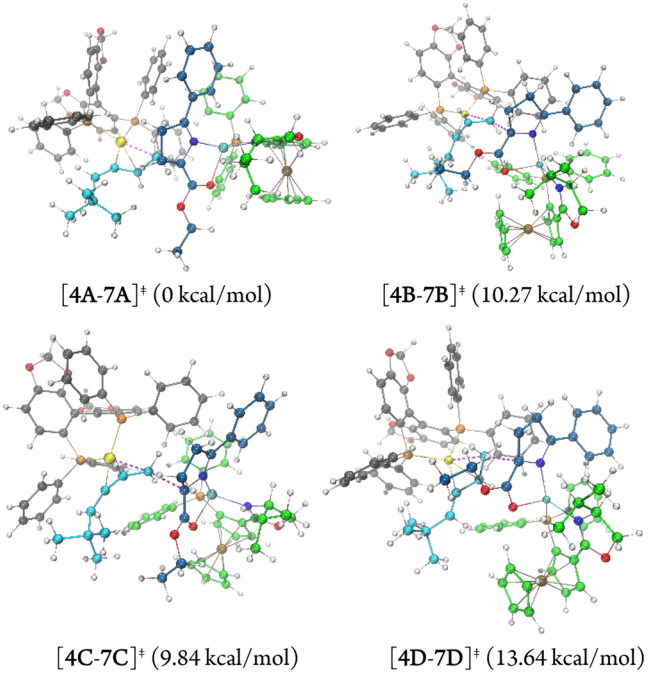
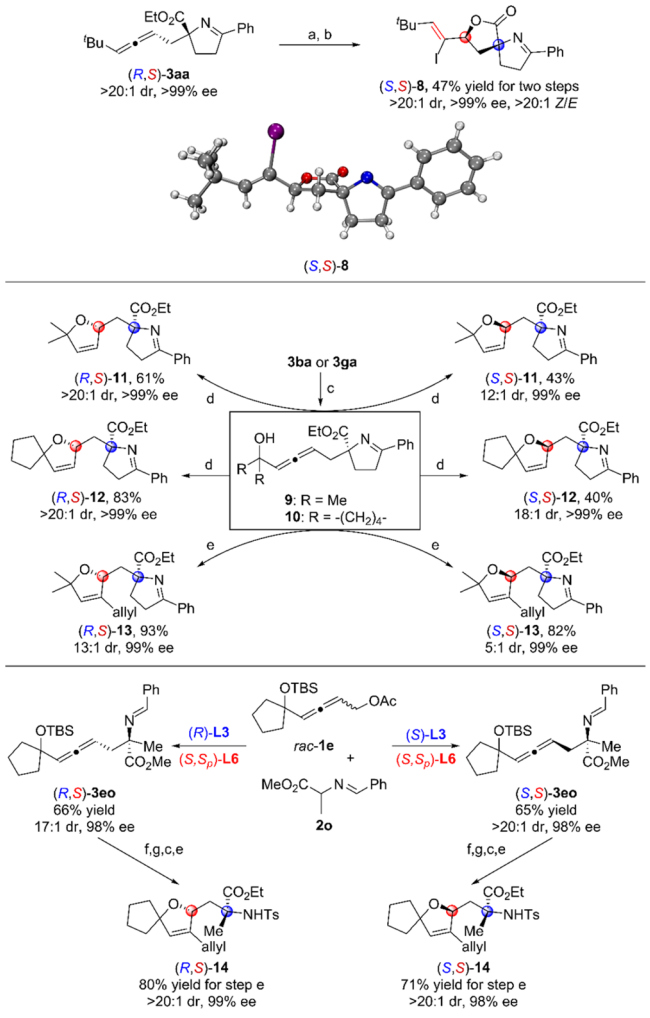


Figure 2. Relative energies of four transition states [**4A**−**7A**]‡, [**4B**−**7B**]‡, [**4C**−**7C**]‡, and [**4D**−**7D**]‡ of α-alkynylidene π-allylpalladium intermediates (**4A**, **4B**, **4C**, and **4D**) with the nucleophile **6**.

To demonstrate the practical applications of this methodology, we devoted our efforts to examining the transformations of the products (Scheme 8). The chiral allene product (*R*,*S*)-

Scheme 8. Representative Transformations of the Chiral Allenylic Products Involving Axial-to-Central Chirality Transfer[a]



[a]Reaction conditions: (a) KOH, EtOH, reflux. (b) I_2, hexane, rt. (c) TBAF, THF, rt. (d) Ph_3PAuCl, AgOTf, CH_2Cl_2, rt. (e) $PdCl_2$, allylic bromide, DMA, rt. (f) 10% citric acid, THF, rt. (g) TsCl, pyridine, rt.

3aa was simply hydrolyzed to the corresponding allenyl acid, which was further converted into the chiral spirocycle product (*S*,*S*)-**8** without any loss of enantioselectivity and diastereoselectivity via electrophilic iodocyclization in the presence of I_2.[25] The structure of the cyclization product was confirmed by X-ray crystallography. Moreover, rearrangement of the deprotected products **9** of chiral allenes **3ba** into trisubstituted dihydrofuran (*R*,*S*)-**11** and (*S*,*S*)-**11** was performed with gold(I) catalysis, providing the products in a diastereodivergent manner with retention of the enantioselectivity.[6d] Analogously, dihydrofuran derivatives (*R*,*S*)-**12** and (*S*,*S*)-**12** can be obtained by rearrangement of the deprotected products **10** from the chiral allenes **3ga**, respectively. To our delight, the Pd-catalyzed cyclization of allenyl alcohol **9** provided the tertsubstituted dihydrofuran (*R*,*S*)-**13** and (*S*,*S*)-**13** in high yields and with high stereoselectivities (13:1 dr and 5:1 dr,

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respectively).[26] Furthermore, amino acids bearing a chiral allene unit, (*R*,*S*)-**3eo** and (*S*,*S*)-**3eo**, could also be converted into dihydrofuran (*R*,*S*)-**14** and (*S*,*S*)-**14** through efficient axial-to-center chirality transfer.

CONCLUSIONS

In summary, we have successfully realized the enantio- and diastereoselective construction of nonadjacent stereocenters in one step through synergistic Pd/Cu catalysis. A series of nonadjacent stereocenters bearing allenyl axial and central chirality could be easily obtained in an enantio- and diastereodivergent manner. Importantly, a broad range of allene substrates, including many examples with less sterically bulky substituents, were also tolerated and delivered the corresponding products in high yields and with high to excellent enantio- and diastereoselectivities. This method represents a rare example of the stereodivergent synthesis of 1,3-nonadjacent stereocenters from the same set of starting materials. Furthermore, the products bearing nonadjacent allenyl axial and central chirality could be applied to the asymmetric synthesis of spirocycle and dihydrofuran structural motifs containing nonadjacent stereogenic centers, which are widely present in numerous important bioactive compounds.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/jacs.1c05087.

Experimental procedures, computational details, and characterization data for all reactions and products, including ^{1}H and ^{13}C NMR spectra, ^{19}F NMR spectra, HPLC spectra, and HRMS (PDF)

Accession Codes

CCDC 2084079−2084081 contain the supplementary crystallographic data for this paper. These data can be obtained free of charge via www.ccdc.cam.ac.uk/data_request/cif, or by emailing data_request@ccdc.cam.ac.uk, or by contacting The Cambridge Crystallographic Data Centre, 12 Union Road, Cambridge CB2 1EZ, UK; fax: +44 1223 336033.

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‖These authors contributed equally.

Notes

The authors declare no competing financial interest.

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This paper was published on August 5, 2021, with an incorrect structure in Scheme 1. The corrected version was reposted on August 9, 2021.

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